Rajdeep Gill

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Education

Bachelor of Science in Computer Engineering (Co-op) May 2026
University of Manitoba *Winnipeg, MB*

- GPA: *4.45/4.5*
- Relevant Coursework: *Data Structures and Algorithms, Systems Engineering Principles, Parallel Programming*

Experience

Machine Learning Research Student May 2024 - Present
Biosensors Research Lab - University of Manitoba *Winnipeg, MB*

- Designed and trained a neural network to predict biological cell viability from cell diffraction patterns, achieving an accuracy of 85%, demonstrating the feasibility of non-invasive cell viability prediction
- Enhanced particle diffraction centroid detection using computer vision techniques, boosting success rate from 75% to 92%, which significantly improved the accuracy of data analysis
- Developed an intuitive data visualization tool to preprocess, filter and display diffraction patterns, facilitating easier data analysis and interpretation
- Integrated neural network and Shapley values into the data visualization tool for deeper insights into model decisions
- Implemented a robust SQL database to efficiently store and manage optical diffraction data, enabling seamless access and analysis for the research team

Data Analysis and Computational Research Student May 2023 - August 2023
Biosensors Research Lab - University of Manitoba *Winnipeg, MB*

- Applied dimensionality reduction techniques (t-SNE, PCA) to compare and cluster datasets, uncovering key correlations and guiding future research efforts
- Accelerated data analysis by 90% through parallel processing, significantly improving workflow efficiency
- Created an oscilloscope interfacing program for a separate lab project, providing easier data collection
- Delivered weekly updates in team meetings, sharing progress and gathering feedback from professors and students

Projects

Discord Clone | *React, TypeScript, HTML/CSS, Next.js*

- Developed a feature-rich Discord Clone with real-time messaging using web sockets
- Utilized Tailwind CSS to create a visually appealing and responsive user interface
- Implemented API services to handle user authentication, voice and video calls
- Provided users with extensive control, allowing them to edit and delete messages, create and delete servers

Microprocessor Communication System | *C, Java*

- Designed and implemented a server using the TCP/IP stack in C, handling multiple concurrent connections
- Developed a Java-based proxy server to interface between an Android application and the microprocessor server
- Designed an Android application to allow users to control the LEDs, monitor temperature and button states

Chess Game | *Python*

- Developed a command-line chess game in Python, with all chess rules successfully implemented
- Implemented multiplayer functionality to enable real-time play between users using sockets
- Allowed users to load and save games to and from a file, using algebraic notation

Awards

Engineering Academic Excellence Award - Computer Engineering 2024

- Awarded to the top student in the Computer Engineering program

NSERC Undergraduate Student Research Award 2023, 2024
President's Scholarship 2021-2024

Technical Skills

Languages: Python, Java, C, C++, JavaScript, HTML/CSS, R, SQL, MATLAB
Frameworks & Libraries: React, TensorFlow, PyTorch, OpenCV, Next.js
Developer Tools: Git, VSCode, Visual Studio, Android Studio, IntelliJ